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eliteigcse.com

Integration: Areas and the Trapezium Rule

Integrate, top limit minus bottom. For an area: sketch, then top curve minus bottom curve. No formula? Trapezium rule.

THE TOPIC ON ONE MAP

Integration: Areas and the Trapezium Rule

01 — Definite integrals

$$[F(x)]_a^b$$

unknown limit k

unknown coefficient

02 — Areas

top - bottom

below the axis

curve and line

03 — Trapezium rule

$$h = \frac{b-a}{n}$$

over or under?

transform the answer

THE TWO TOOLS

$$\int_a^b f(x) dx = F(b) - F(a) \quad \int_a^b y dx \approx \frac{h}{2} \{y_0 + y_n + 2(y_1 + \dots + y_{n-1})\}$$

$h = \frac{b-a}{n}$ · n strips = $n+1$ ordinates (y values) · rewrite roots and fractions as powers first

01 Definite Integrals: Limits and Unknown Constants

Rewrite every term as ax^n , integrate (no $+c$), substitute the top limit and subtract the bottom. If a limit or a coefficient is unknown, the integral equals a number: that is your equation.

WHEN YOU SEE → YOU DO

“use calculus to find the value of

$$\int_4^9 \dots dx$$

→ rewrite, integrate, substitute

“given $\int_k^{2k} \dots dx = 14$, find k ”

→ an equation in k

“($x+2$) is a factor and $\int_3^5 f(x) dx = 176$ ”

→ two equations, solve for A, B

$$\frac{(x-k)^2}{\sqrt{x}}$$

→ expand the top, divide each term by $x^{\frac{1}{2}}$

WORKED MODEL · JANUARY 2026

$$\int_k^{2k} \left(\frac{12}{x^2} + 4 \right) dx = 14, k > 0. \text{ Find } k.$$

$$1 \quad \left[-\frac{12}{x} + 4x \right]_k^{2k} = \frac{6}{k} + 4k$$

integrate, substitute

$$2 \quad \frac{6}{k} + 4k = 14 \Rightarrow 2k^2 - 7k + 3 = 0$$

equation

$$\text{ANSWER} \quad k = \frac{1}{2} \text{ or } k = 3$$

IDEA 01

Practise: Definite Integrals: Limits and Unknown Constants

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Evaluate

P2 · TOPIC 12 3 marks

Evaluate $\int_1^4 \left(3\sqrt{x} - \frac{2}{x^2} \right) dx$.

Q2 DR ESLAM Find the limit

P2 · TOPIC 12 4 marks

Given $\int_1^k (2x + 1) dx = 18$, $k > 1$, find k .

Integrate, substitute, then solve the quadratic in k .

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P2 · TOPIC 12 4 marks

1 Evaluate

[i] $\int_1^2 4x \, dx,$

[ii] $\int_0^3 (9x^2 + 4x) \, dx.$

(4 marks)

Q4**PRACTICE**

Practice · Hard

P2 · TOPIC 12**5 marks**

- 1 Use calculus to find the value of

$$\int_4^9 \frac{x^2 + 1}{\sqrt{x}} dx$$

(5 marks)

Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 12 5 marks

2 Given

$$\int_q^{4q} 5x\sqrt{x} \, dx = 15\,066$$

find the value of the constant q .**(5 marks)**

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · June 2022 · Q7

P2 · TOPIC 12 7 marks

7. $f(x) = Ax^3 + 6x^2 - 4x + B$

where A and B are constants.

Given that

- $(x + 2)$ is a factor of $f(x)$

- $\int_3^5 f(x) dx = 176$

find the value of A and the value of B .

(7)

(Total 7 marks)

Q7**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q4

P2 · TOPIC 12 5 marks

4. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

Given that k is a positive constant and

$$\int_k^{2k} \left(\frac{12}{x^2} + 4 \right) dx = 14$$

find the possible values of k .

(5)

(Total for Question 4 is 5 marks)

02 Areas: Curves, Lines and Below the Axis

Sketch and find where the curves meet — those are the limits. The area between two graphs is $\int(\text{top} - \text{bottom})dx$. An integral below the x-axis is negative: find each part separately and add the sizes.

WHEN YOU SEE → YOU DO

“the region bounded by C and the line l ”

→ meeting points, then top - bottom

“bounded by the curve, the y-axis and $y = 17$ ”

→ rectangle - area under the curve

a region part above, part below the axis

→ split at the root, add the sizes

“area of R_1 equals area of R_2 ”

→ the signed integral is 0

“use algebraic integration to find the exact area”

→ exact: fractions, surds, π

a triangle or a trapezium in the region

→ use the shape formula, then add or subtract

Sketch

Meeting points

Top - bottom

Integrate and add

WORKED MODEL · JUNE 2025

$y = 3x + \frac{16}{x^2} - 8$; $P(1, 11)$ and $Q(4, 5)$. Area between C and the segment PQ .

1 trapezium under PQ : $\frac{1}{2}(11 + 5)(3) = 24$

the line

2 under C : $\left[\frac{3}{2}x^2 - \frac{16}{x} - 8x\right]_1^4 = \frac{21}{2}$

the curve

ANSWER $24 - \frac{21}{2} = \frac{27}{2}$

IDEA 02

Practise: Areas: Curves, Lines and Below the Axis

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM Curve and line

P2 · TOPIC 12 5 marks

Find the area enclosed by $y = x^2 - 4x + 6$ and $y = x + 2$.

Q9 DR ESLAM Below the axis

P2 · TOPIC 12 3 marks

Find the area between $y = x^2 - 4$ and the x-axis.

SELECTED PRACTICE

Q10**PRACTICE**

Practice · Easy

P2 · TOPIC 12 5 marks

- 3 (a)** The area bounded by the curve with equation $y = 9 - x^2$, the x -axis and the vertical lines with equations $x = 1$ and $x = 2$ is to be found.

Write down an integral that would find this area.

(2 marks)

- (b)** Evaluate your integral from part [a] and hence find the area described above.

(3 marks)

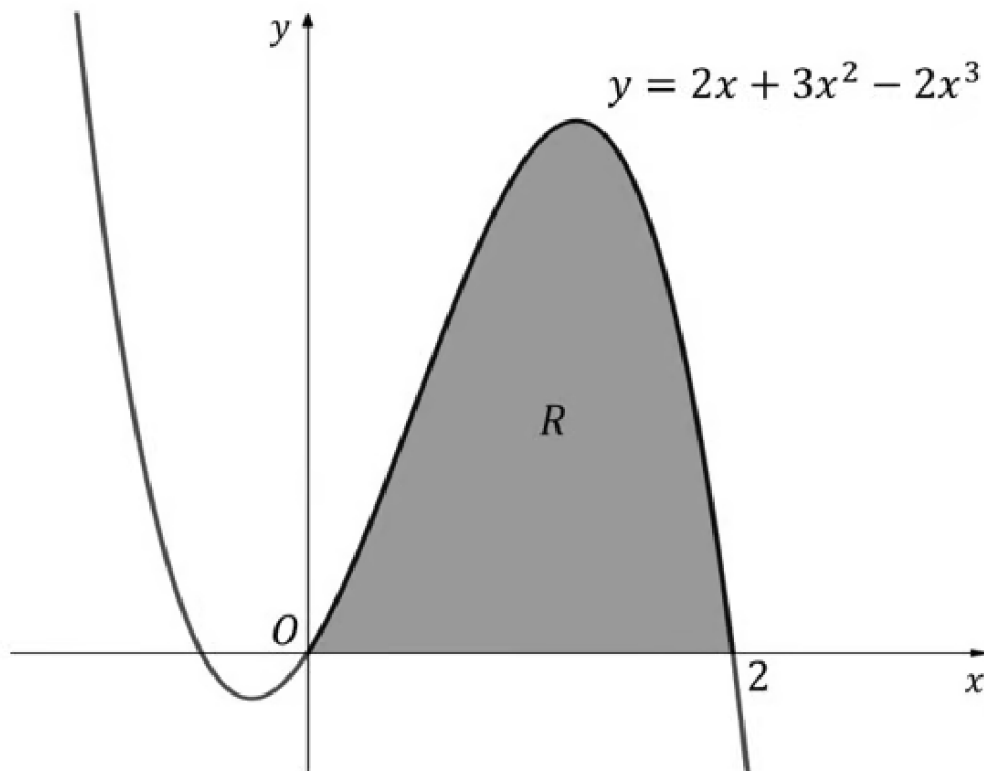
Q11**PRACTICE**

Practice · Hard

P2 · TOPIC 12

4 marks

- 3 The diagram below shows part of the graph of $y = 2x + 3x^2 - 2x^3$.



Find the area of the shaded region labelled R .

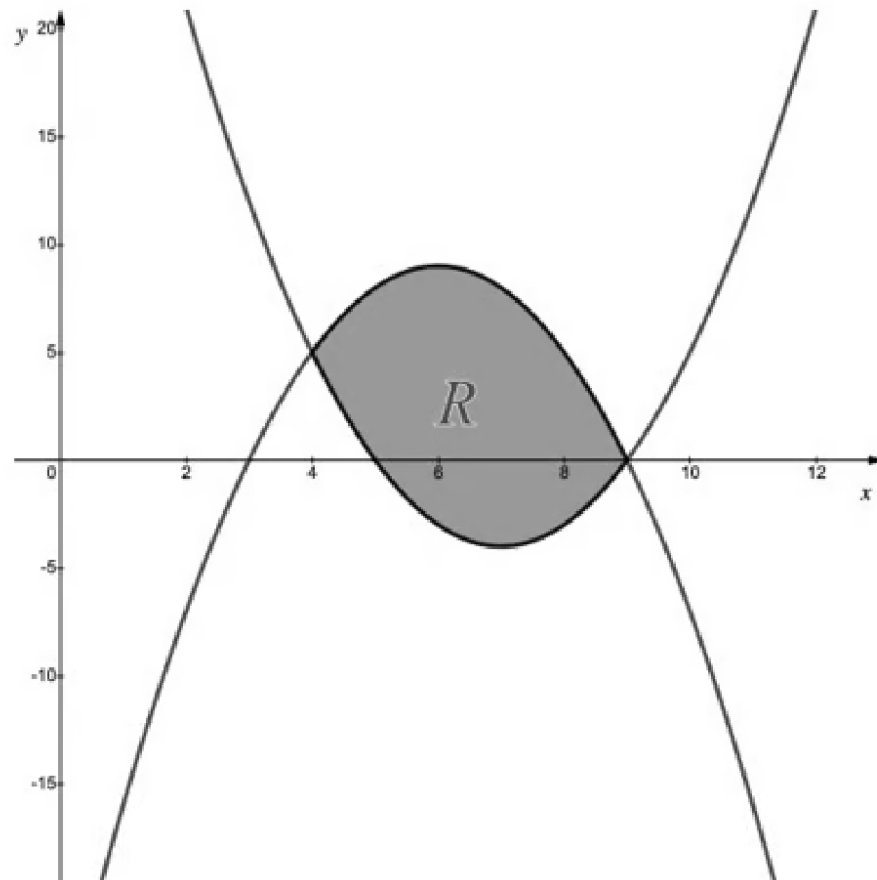
(4 marks)

Q12**PRACTICE**

Practice · Very Hard

P2 · TOPIC 12**8 marks**

7 The diagram below shows the graphs of $y = 12x - x^2 - 27$ and $y = x^2 - 14x + 45$.



Find the area of the shaded region, R .

(8 marks)

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA12 · October 2019 · Q8

P2 · TOPIC 12 9 marks

8. Solutions relying on calculator technology are not acceptable in this question.

(i)

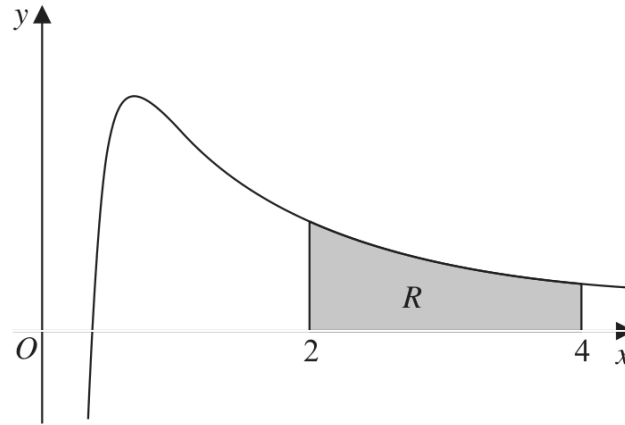
**Figure 2**

Figure 2 shows a sketch of part of a curve with equation

$$y = \frac{8\sqrt{x} - 5}{2x^2} \quad x > 0$$

The region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 4$

Find the exact area of R .

(5)

(ii) Find the value of the constant k such that

$$\int_{-3}^6 \left(\frac{1}{2}x^2 + k \right) dx = 55$$

(4)

(Total 9 marks)

Q14 PAST PAPER Pearson IAL WMA12 · January 2020 · Q4

P2 · TOPIC 12 6 marks

4.

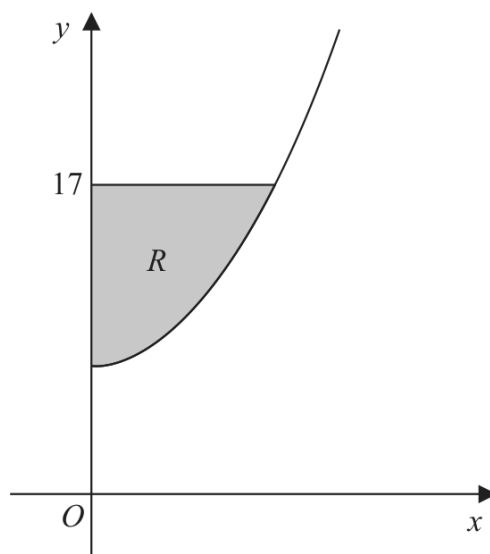
**Figure 1**

Figure 1 shows a sketch of the curve with equation

$$y = 2x^2 + 7 \quad x \geq 0$$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the y -axis and the line with equation $y = 17$

Find the exact area of R .

(6)**(Total 6 marks)**

Q15**PAST PAPER**

Pearson IAL WMA12 · January 2022 · Q9

P2 · TOPIC 12 10 marks

9.

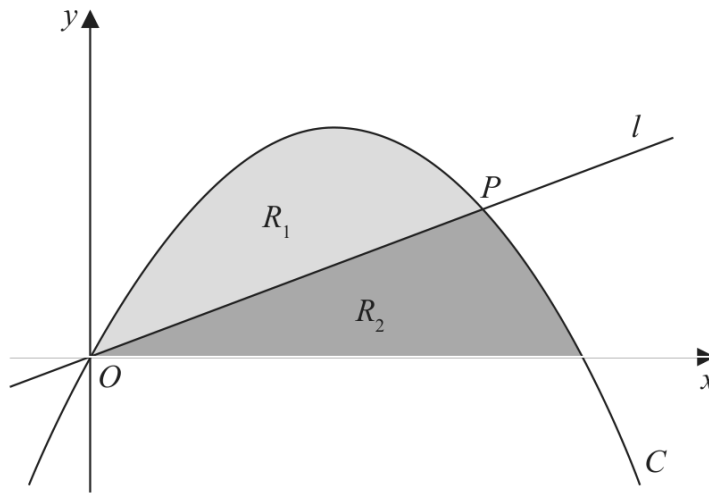
**Figure 2**

Figure 2 shows

- the curve C with equation $y = x - x^2$
- the line l with equation $y = mx$, where m is a constant and $0 < m < 1$

The line and the curve intersect at the origin O and at the point P .

- (a) Find, in terms of
- m
- , the coordinates of
- P
- .

(2)The region R_1 , shown shaded in Figure 2, is bounded by C and l .

- (b) Show that the area of
- R_1
- is

$$\frac{(1-m)^3}{6}$$

(5)The region R_2 , also shown shaded in Figure 2, is bounded by C , the x -axis and l .Given that the area of R_1 is equal to the area of R_2

- (c) find the exact value of
- m
- .

(3)**(Total 10 marks)**

Q16**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q9

P2 · TOPIC 12**8 marks**

9. In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

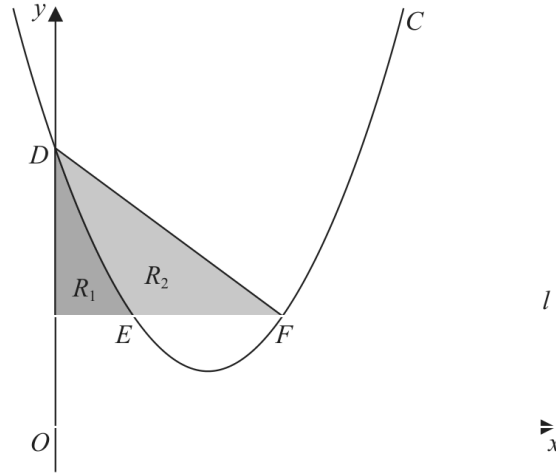


Figure 3

Figure 3 shows

- the curve C with equation $y = x^2 - 4x + 5$
- the line l with equation $y = 2$

The curve C intersects the y -axis at the point D .

(a) Write down the coordinates of D .

(1)

The curve C intersects the line l at the points E and F , as shown in Figure 3.

(b) Find the x coordinate of E and the x coordinate of F .

(2)

Shown shaded in Figure 3 is

- the region R_1 which is bounded by C , l and the y -axis
- the region R_2 which is bounded by C and the line segments EF and DF

Given that $\frac{\text{area of } R_1}{\text{area of } R_2} = k$, where k is a constant,

(c) use algebraic integration to find the exact value of k , giving your answer as a simplified fraction.

(5)

(Total for Question 9 is 8 marks)

Q17**PAST PAPER**

Pearson IAL WMA12 · June 2023 · Q10

P2 · TOPIC 12 10 marks

10. The curve C has equation

$$y = \frac{(x - k)^2}{\sqrt{x}} \quad x > 0$$

where k is a **positive** constant.

(a) Show that

$$\int_1^{16} \frac{(x - k)^2}{\sqrt{x}} \, dx = ak^2 + bk + \frac{2046}{5}$$

where a and b are integers to be found.

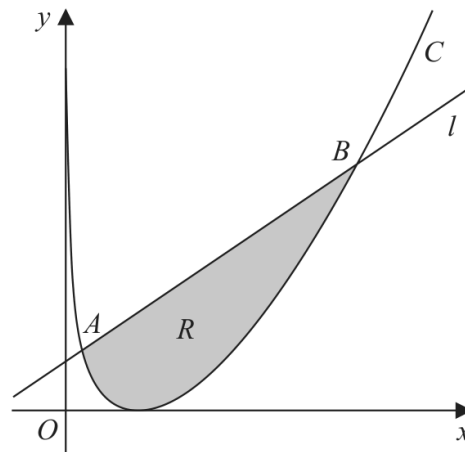
(5)

Figure 1

Figure 1 shows a sketch of the curve C and the line l .

Given that l intersects C at the point $A(1, 9)$ and at the point $B(16, q)$ where q is a constant,

(b) show that $k = 4$

(2)

The region R , shown shaded in Figure 1, is bounded by C and l

Using the answers to parts (a) and (b),

(c) find the area of region R

(3)

(Total for Question 10 is 10 marks)

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

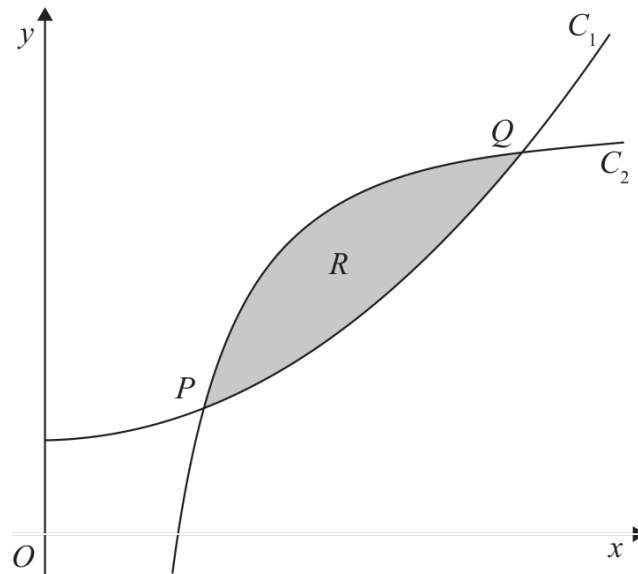


Figure 1

Figure 1 shows a sketch of part of the curve C_1 with equation

$$y = x^2 + 3 \quad x > 0$$

and part of the curve C_2 with equation

$$y = 13 - \frac{9}{x^2} \quad x > 0$$

The curves C_1 and C_2 intersect at the points P and Q as shown in Figure 1.

(a) Use algebra to find the x coordinate of P and the x coordinate of Q .

(4)

The finite region R , shown shaded in Figure 1, is bounded by C_1 and C_2

(b) Use algebraic integration to find the exact area of R .

(4)

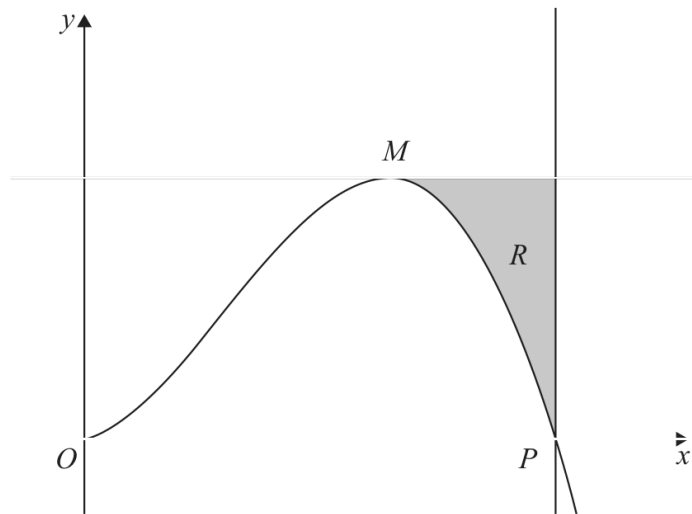
(Total for Question 8 is 8 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · January 2025 · Q9

P2 · TOPIC 12 12 marks

9.

**Figure 3**

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 3 shows a sketch of part of the curve with equation

$$y = \frac{9x^2(5 - \sqrt{x})}{5} \quad x \geq 0$$

The curve has a turning point at the point M , as shown in Figure 3.

(a) Using calculus, find the coordinates of M .

(5)

The curve crosses the x -axis at the point P , as shown in Figure 3.

(b) Use algebra to find the x coordinate of P .

(2)

The finite region R , shown shaded in Figure 3, is bounded by the curve, the line through M parallel to the x -axis and the line through P parallel to the y -axis.

(c) Use algebraic integration to find the area of R , giving your answer to one decimal place.

(5)

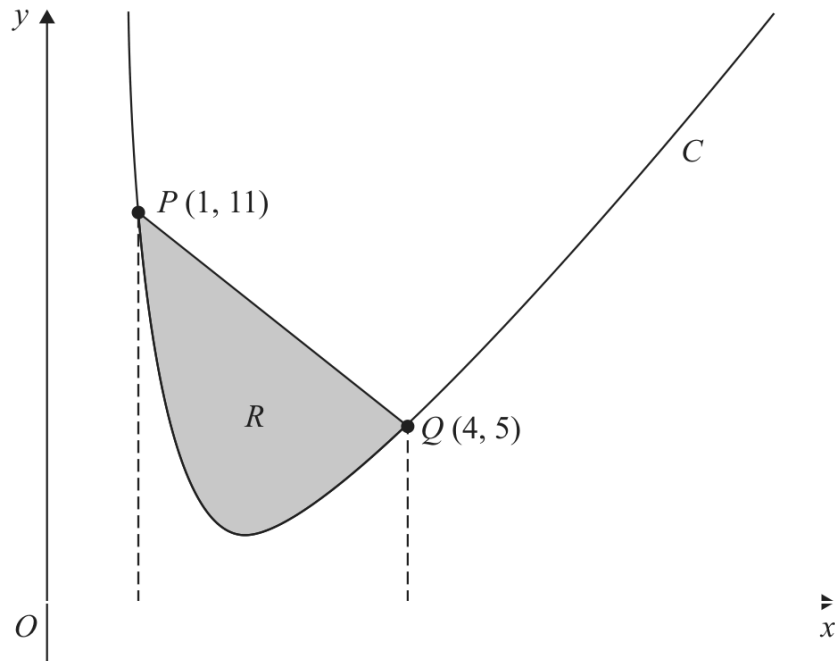
(Total for Question 9 is 12 marks)

Q20**PAST PAPER**

Pearson IAL WMA12 · June 2025 · Q3

P2 · TOPIC 12**5 marks**

3:

**Figure 1**

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 1 shows a sketch of the curve C with equation

$$y = 3x + \frac{16}{x^2} - 8 \quad x > 0$$

The points $P(1, 11)$ and $Q(4, 5)$ lie on C and are shown in Figure 1.

The region R , shown shaded in Figure 1, is bounded by C and line segment PQ .

Use algebraic integration to find the area of R .

(5)

(Total for Question 3 is 5 marks)

7.

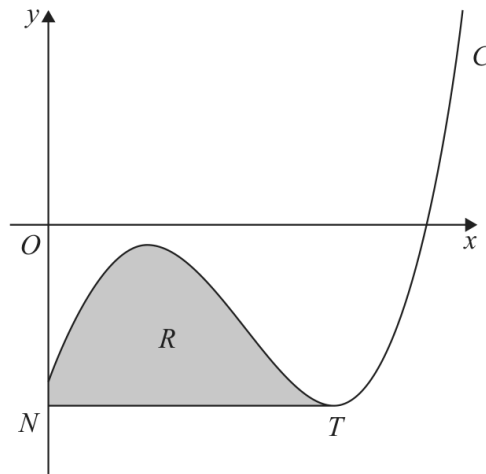


Figure 3

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 3 shows a sketch of part of the curve C with equation

$$y = x^3 - 4x^{\frac{5}{2}} - kx^{\frac{1}{2}} + 28x - 44 \quad x \geq 0$$

where k is a positive constant.

- (a) Find $\frac{dy}{dx}$ in simplest form. (2)

The point T , shown in Figure 3, is a minimum stationary point on C .

Given that the x coordinate of T is 9

- (b) show that $k = 6$ (2)

The line through T parallel to the x -axis meets the y -axis at the point N .

The finite region R , shown shaded in Figure 3, is bounded by C , the y -axis and the line segment NT .

- (c) Use algebraic integration to find the area of R , giving the answer to 3 significant figures. (6)

(Total for Question 7 is 10 marks)

8.

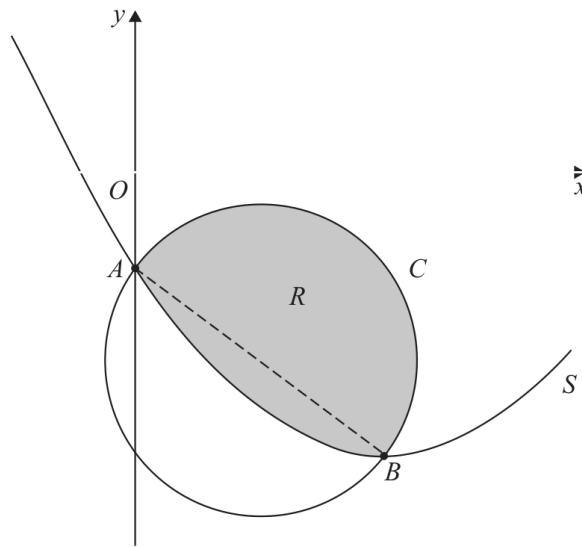


Figure 3

Figure 3 shows a sketch of a circle C and a parabola S .

The circle C has equation $x^2 + y^2 - 8x + 12y + 27 = 0$

(a) Find

- (i) the coordinates of the centre of C
- (ii) the radius of C

(3)

The parabola S

- has equation $y = ax^2 - \frac{3}{2}x - 3$
- intersects C on the y -axis at the point $A(0, -3)$
- intersects C again at the point B where AB is a diameter of C

Given that B is the minimum point of S ,

(b) show that $a = \frac{3}{32}$

(3)

The finite region R is bounded by S and C and is shown shaded in Figure 3.

(c) Use algebraic integration to find the exact area of R .

(7)

(Total for Question 8 is 13 marks)

03 The Trapezium Rule: Estimate, Then Transform

Find h , list every y value (complete the table first), and use the formula with the first and last y once and all the middle ones twice. Then use the answer: a constant multiple multiplies the answer, and $+k$ adds $k \times$ the width.

WHEN YOU SEE → YOU DO

“complete the table”

→ substitute each x — to the accuracy asked

“using all the y values”

→ h is the gap between the x values

$$\int_a^b (3f(x) - 2) dx$$

→ $3 \times \text{estimate} - 2(b - a)$

$\log_2(4x^2) = 2 + 2 \log_2 x$ **style**

→ split with log laws into the known integral

“overestimate or underestimate?”

→ curve bends up (convex) → over;
bends down → under

“how could the estimate be improved?”

→ more strips (smaller h)

WORKED MODEL · JANUARY 2023

$x = -1, -0.5, 0, 0.5, 1$; $y = 2.287, 4.470, 6.719, 7.291, 2.834$. Estimate $\int_{-1}^1 f(x) dx$.

1 $h = 0.5$

gap

2 $\frac{0.5}{2} \{2.287 + 2.834 + 2(4.470 + 6.719 + 7.291)\}$

ends once, middle twice

ANSWER ≈ 10.52 ; $\int_{-1}^1 \{f(x) - 2\} dx \approx 10.52 - 4 = 6.52$

× **TRAP** Using $n + 1$ for the number of strips. Five y values means four strips.

IDEA 03

Practise: The Trapezium Rule: Estimate, Then Transform

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM Trapezium rule

P2 · TOPIC 12 3 marks

Use the trapezium rule with 4 strips to estimate $\int_0^2 \sqrt{1+x^2} dx$.

Q24 DR ESLAM Use the estimate

P2 · TOPIC 12 2 marks

Given $\int_0^2 f(x) dx \approx 5.4$, estimate $\int_0^2 \{3f(x) + 2\} dx$.

SELECTED PRACTICE

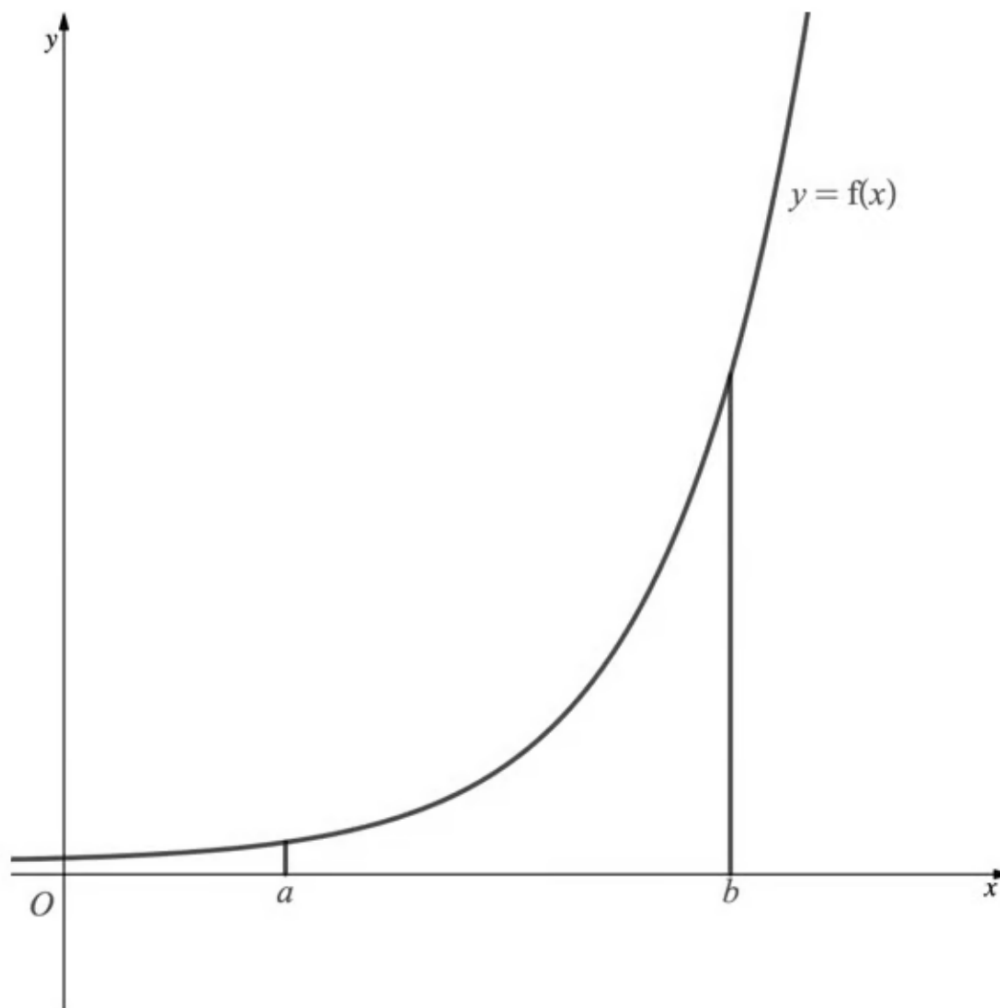
Q25**PRACTICE**

Practice · Easy

P2 · TOPIC 12 2 marks

- 7 A student is estimating the area bounded by the curve $y = f(x)$, the x -axis and the lines $x = a$ and $x = b$.

The student intends to estimate the area by using trapezia of equal width.



Add to the diagram above to show how the student can use 4 trapezia to estimate the area.

(2 marks)

Q26**PRACTICE**

Practice · Medium

P2 · TOPIC 12 8 marks**11 (a)** The trapezium rule is to be used to estimate the integral

$$\int_{-2}^0 x^3 + 8 \, dx$$

By completing the table of values below, use the trapezium rule to estimate the integral given above.

x	-2	-1.5	-1	-0.5	0
$y = x^3 + 8$					

(5 marks)

(b) i] By finding the exact value of the integral above, find the percentage error of your estimate from part [a].

ii] Explain how you could increase the accuracy of your trapezium rule estimate.

(3 marks)

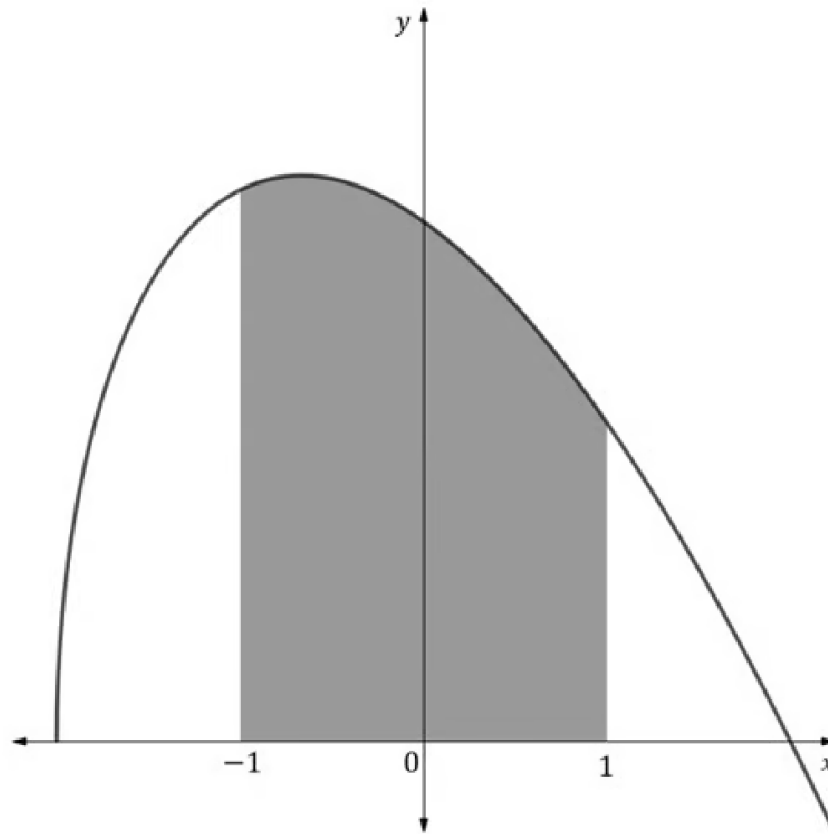
Q27**PRACTICE**

Practice · Very Hard

P2 · TOPIC 12 6 marks

9 (a) The diagram below shows the graph of

$$y = \frac{4 - x^2}{\sqrt{x + 2}}, \quad x > -2.$$



Use the trapezium rule with six strips to find an estimate of the integral

$$\int_{-1}^1 \left(\frac{4 - x^2}{\sqrt{x + 2}} \right) dx$$

to five significant figures.

(4 marks)

(b) By using your calculator to find the exact value of the integral to five significant figures, find the percentage error of your estimate from part [a].

(2 marks)

PAST PAPERS

Q28 PAST PAPER Pearson IAL WMA12 · October 2019 · Q5

P2 · TOPIC 12 7 marks

5. (a) Given
- $0 < a < 1$
- , sketch the curve with equation

$$y = a^x$$

showing the coordinates of the point at which the curve crosses the y-axis.

(2)

x	2	2.5	3	3.5	4
y	4.25	6.427	9.125	12.34	16.06

The table above shows corresponding values of x and y for $y = x^2 + \left(\frac{1}{2}\right)^x$

The values of y are given to 4 significant figures as appropriate.

Using the trapezium rule with all the values of y in the given table,

- (b) obtain an estimate for $\int_2^4 \left(x^2 + \left(\frac{1}{2}\right)^x \right) dx$

(3)

Using your answer to part (b) and making your method clear, estimate

- (c) $\int_2^4 \left(x(x-3) + \left(\frac{1}{2}\right)^x \right) dx$

(2)

(Total 7 marks)

Q29**PAST PAPER**

Pearson IAL WMA12 · January 2020 · Q1

P2 · TOPIC 12 7 marks

1. The table below shows corresponding values of x and y for $y = \log_2(2x)$

The values of y are given to 2 decimal places as appropriate.

x	2	5	8	11	14
y	2	3.32	4	4.46	4.81

Using the trapezium rule with all the values of y in the given table,

- (a) obtain an estimate for $\int_2^{14} \log_2(2x) dx$, giving your answer to one decimal place. (3)

Using your answer to part (a) and making your method clear, estimate

- (b) (i) $\int_2^{14} \frac{\log_2(4x^2)}{5} dx$
 (ii) $\int_2^{14} \log_2\left(\frac{2}{x}\right) dx$ (4)

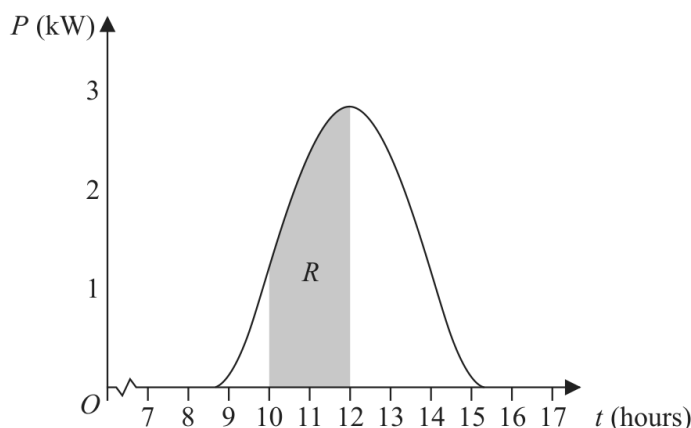
(Total 7 marks)

Q30**PAST PAPER**

Pearson IAL WMA12 · January 2021 · Q7

P2 · TOPIC 12**6 marks**

7.

**Figure 1**

Solar panels are installed on the roof of a building.

The power, P , produced on a particular day, in kW, can be modelled by the equation

$$P = 0.95 + 2^{t-12} + 2^{12-t} - (t-12)^2 \quad 8.5 \leq t \leq 15.2$$

where t is the time in hours after midnight. The graph of P against t is shown in Figure 1.

A table of values of t and P is shown below, with the values of P given to 4 significant figures where appropriate.

Time, t (hours)	10	10.5	11	11.5	12
Power, P (kW)		1.882	2.45		2.95

- (a) Use the given equation to complete the table, giving the values of P to 4 significant figures where appropriate.

(2)

The amount of energy, in kWh, produced between 10:00 and 12:00 can be found by calculating the area of region R , shown shaded in Figure 1.

- (b) Use the trapezium rule, with all the values of P in the completed table, to find an estimate for the amount of energy produced between 10:00 and 12:00. Give your answer to 2 decimal places.

(4)**(Total 6 marks)**

Q31 PAST PAPER Pearson IAL WMA12 · June 2021 · Q5

P2 · TOPIC 12 10 marks

5.

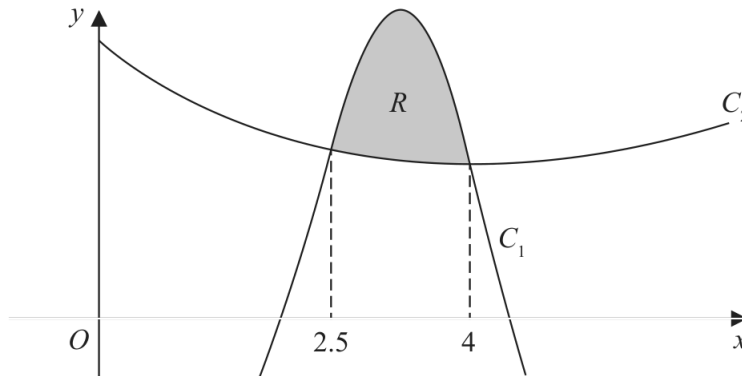
**Figure 2**

Figure 2 shows a sketch of part of the graph of the curves C_1 and C_2

The curves intersect when $x = 2.5$ and when $x = 4$

A table of values for some points on the curve C_1 is shown below, with y values given to 3 decimal places as appropriate.

x	2.5	2.75	3	3.25	3.5	3.75	4
y	5.453	7.764	9.375	9.964	9.367	7.626	5

Using the trapezium rule with all the values of y in the table,

- (a) find, to 2 decimal places, an estimate for the area bounded by the curve C_1 , the line with equation $x = 2.5$, the x -axis and the line with equation $x = 4$
- (4)

The curve C_2 has equation

$$y = x^{\frac{3}{2}} - 3x + 9 \quad x > 0$$

- (b) Find $\int \left(x^{\frac{3}{2}} - 3x + 9 \right) dx$
- (3)

The region R , shown shaded in Figure 2, is bounded by the curves C_1 and C_2

- (c) Use the answers to part (a) and part (b) to find, to one decimal place, an estimate for the area of the region R .
- (3)

(Total 10 marks)

Q32**PAST PAPER**

Pearson IAL WMA12 · October 2021 · Q3

P2 · TOPIC 12**8 marks**

3.

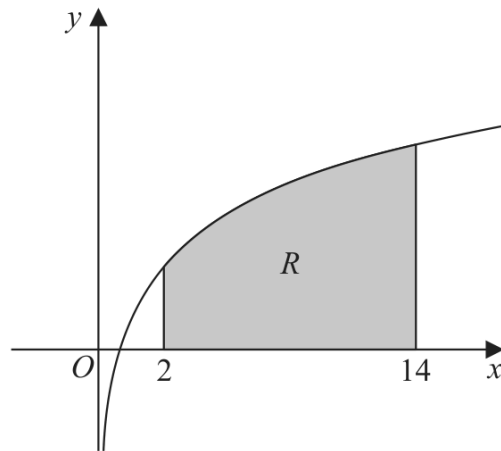
**Figure 1**

Figure 1 shows a sketch of part of the curve with equation $y = \log_{10} x$

The region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 14$

Using the trapezium rule with four strips of equal width,

(a) show that the area of R is approximately 10.10 (3)

(b) Explain how the trapezium rule could be used to obtain a more accurate estimate for the area of R . (1)

(c) Using the answer to part (a) and making your method clear, estimate the value of

(i) $\int_2^{14} \log_{10} \sqrt{x} \, dx$

(ii) $\int_2^{14} \log_{10} 100x^3 \, dx$

(4)**(Total 8 marks)**

Q33**PAST PAPER**

Pearson IAL WMA12 · January 2022 · Q1

P2 · TOPIC 12 7 marks

1. The table below shows corresponding values of x and y for

$$y = 2^{5-\sqrt{x}}$$

The values of y are given to 3 decimal places.

x	5	5.5	6	6.5	7
y	6.792	6.298	5.858	5.466	5.113

Using the trapezium rule with all the values of y in the given table,

- (a) obtain an estimate for

$$\int_5^7 2^{5-\sqrt{x}} dx$$

giving your answer to 2 decimal places.

(3)

- (b) Using your answer to part (a) and making your method clear, estimate

(i) $\int_5^7 2^{6-\sqrt{x}} dx$

(ii) $\int_5^7 (3 + 2^{5-\sqrt{x}}) dx$

(4)

(Total 7 marks)

Q34 PAST PAPER Pearson IAL WMA12 · June 2022 · Q2

P2 · TOPIC 12 8 marks

2.

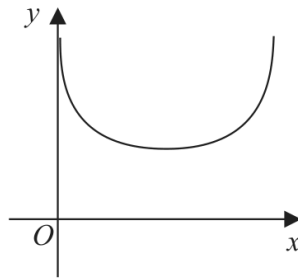
**Figure 1**

Figure 1 shows the graph of

$$y = 1 - \log_{10}(\sin x) \quad 0 < x < \pi$$

where x is in radians.The table below shows some values of x and y for this graph, with values of y given to 3 decimal places.

x	0.5	1	1.5	2	2.5	3
y	1.319		1.001		1.223	1.850

(a) Complete the table above, giving values of y to 3 decimal places.

(2)

(b) Use the trapezium rule with all the y values in the completed table to find, to 2 decimal places, an estimate for

$$\int_{0.5}^3 (1 - \log_{10}(\sin x)) dx$$

(3)

(c) Use your answer to part (b) to find an estimate for

$$\int_{0.5}^3 (3 + \log_{10}(\sin x)) dx$$

(3)

(Total 8 marks)

Q35 PAST PAPER Pearson IAL WMA12 · October 2022 · Q6

P2 · TOPIC 12 7 marks

6. The curve C_1 has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below, with the y values rounded to 4 decimal places where appropriate.

x	0	0.5	1	1.5	2
y	3	2.6833	2.4	2.1466	1.92

- (a) Use the trapezium rule with all the values of y in the table to find an approximation for

$$\int_0^2 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

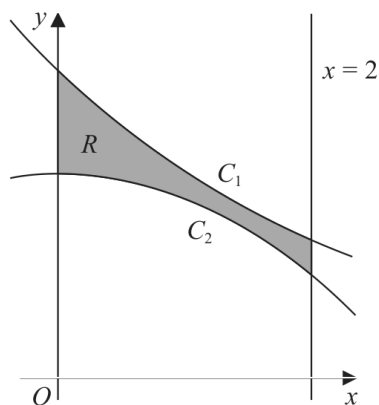


Figure 1

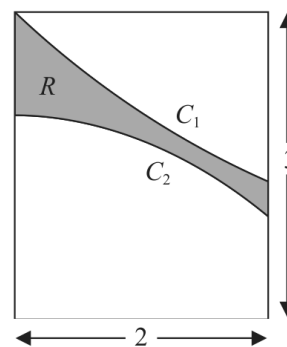


Figure 2

The region R , shown shaded in Figure 1, is bounded by

- the curve C_1
- the curve C_2 with equation $y = 2 - \frac{1}{4}x^2$
- the line with equation $x = 2$
- the y -axis

The region R forms part of the design for a logo shown in Figure 2.

The design consists of the shaded region R inside a rectangle of width 2 and height 3

Using calculus and the answer to part (a),

- (b) calculate an estimate for the percentage of the logo which is shaded.

(4)

(Total for Question 6 is 7 marks)

Q36**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q1

P2 · TOPIC 12 6 marks

1.

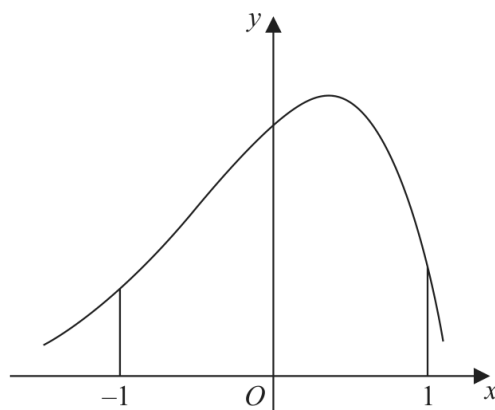
**Figure 1**

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$

The table below shows some corresponding values of x and y for this curve.

The values of y are given to 3 decimal places.

x	-1	-0.5	0	0.5	1
y	2.287	4.470	6.719	7.291	2.834

Using the trapezium rule with all the values of y in the given table,

(a) obtain an estimate for

$$\int_{-1}^1 f(x) \, dx$$

giving your answer to 2 decimal places.

(3)

(b) Use your answer to part (a) to estimate

(i) $\int_{-1}^1 (f(x) - 2) \, dx$

(ii) $\int_1^3 f(x-2) \, dx$

(3)

(Total for Question 1 is 6 marks)

Q37 PAST PAPER Pearson IAL WMA12 · June 2023 · Q1

P2 · TOPIC 12 3 marks

1. The continuous curve C has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	4.0	4.2	4.4	4.6	4.8	5.0
y	9.2	8.4556	3.8512	5.0342	7.8297	8.6

Use the trapezium rule with all the values of y in the table to find an approximation for

$$\int_4^5 f(x) \, dx$$

giving your answer to 3 decimal places.

(3)

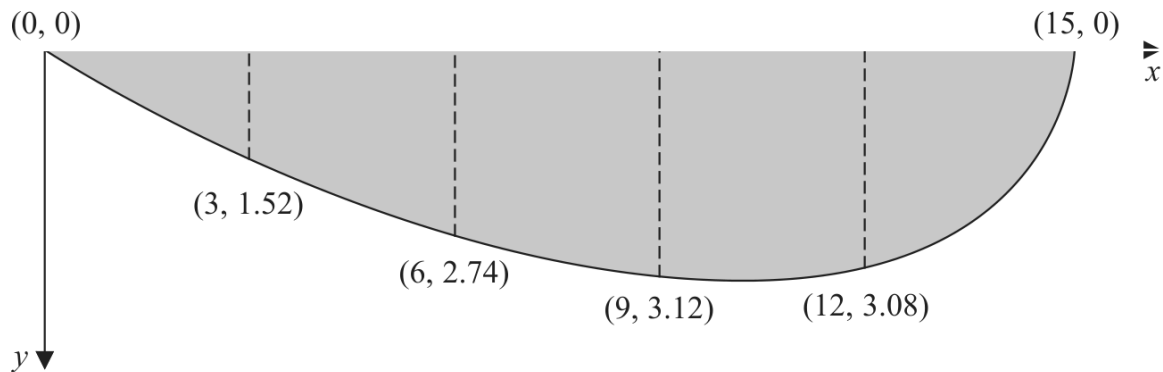
(Total for Question 1 is 3 marks)

Q38**PAST PAPER**

Pearson IAL WMA12 · October 2023 · Q6

P2 · TOPIC 12**6 marks**

6.

**Figure 1**

A river is being studied.

At one particular place, the river is 15 m wide.

The depth, y metres, of the river is measured at a point x metres from one side of the river.

Figure 1 shows a plot of the cross-section of the river and the coordinate values (x, y)

- (a) Use the trapezium rule with all the y values given in Figure 1 to estimate the cross-sectional area of the river.

(3)

The water in the river is modelled as flowing at a constant speed of 1.5 m s^{-1} across the whole of the cross-section.

- (b) Use the model and the answer to part (a) to estimate the volume of water flowing through this section of the river each minute, giving your answer in m^3 to 2 significant figures.

(2)

Assuming the model,

- (c) state, giving a reason for your answer, whether your answer for part (b) is an overestimate or an underestimate of the true volume of water flowing through this section of the river each minute.

(1)

(Total for Question 6 is 6 marks)

Q39**PAST PAPER**

Pearson IAL WMA12 · January 2024 · Q4

P2 · TOPIC 12 9 marks

4. (a) Sketch the curve with equation

$$y = a^{-x} + 4$$

where a is a constant and $a > 1$

On your sketch show

- the coordinates of the point of intersection of the curve with the y -axis
- the equation of the asymptote to the curve.

(3)

x	-4	-1.5	1	3.5	6	8.5
y	13	6.280	4.577	4.146	4.037	4.009

The table above shows corresponding values of x and y for $y = 3^{-\frac{1}{2}x} + 4$

The values of y are given to four significant figures, as appropriate.

Using the trapezium rule with all the values of y in the table,

- (b) find an approximate value for

$$\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx$$

giving your answer to two significant figures.

(3)

- (c) Using the answer to part (b), find an approximate value for

(i) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} \right) dx$

(ii) $\int_{-4}^{8.5} \left(3^{-\frac{1}{2}x} + 4 \right) dx + \int_{-8.5}^4 \left(3^{\frac{1}{2}x} + 4 \right) dx$

(3)

(Total for Question 4 is 9 marks)

Q40**PAST PAPER**

Pearson IAL WMA12 · June 2024 · Q6

P2 · TOPIC 12**9 marks**

6. (a) Sketch the curve with equation

$$y = a^x + 4$$

where a is a positive constant greater than 1

On your sketch, show

- the coordinates of the point of intersection of the curve with the y -axis
- the equation of the asymptote of the curve

(3)

x	2	2.3	2.6	2.9	3.2	3.5
y	0	0.3246	0.8629	1.6643	2.7896	4.3137

The table shows corresponding values of x and y for

$$y = 2^x - 2x$$

with the values of y given to 4 decimal places as appropriate.Using the trapezium rule with all the values of y in the given table,(b) obtain an estimate for $\int_2^{3.5} (2^x - 2x) dx$, giving your answer to 2 decimal places.**(3)**

(c) Using your answer to part (b) and making your method clear, estimate

(i) $\int_2^{3.5} (2^x + 2x) dx$

(ii) $\int_2^{3.5} (2^{x+1} - 4x) dx$

(3)**(Total for Question 6 is 9 marks)**

Q41**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q2

P2 · TOPIC 12 5 marks

2. The table shows corresponding values of x and y for a continuous curve with equation $y = f(x)$ between $x = -4$ and $x = 5$, where a is a constant.

x	-4	-2.5	-1	0.5	2	3.5	5
y	4.16	2.91	a	1.73	1.37	1.43	2.28

The trapezium rule is used with all the y values in the table to find an approximation for

$$\int_{-4}^5 f(x) \, dx$$

Given that the value of this approximation is 19.3

- (a) find the value of the constant a to 3 significant figures.

(3)

- (b) Use the given answer of 19.3 to find an approximate value for

$$\int_{-4}^5 (2f(x) - 3) \, dx$$

(2)

(Total for Question 2 is 5 marks)

Q42**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q1

P2 · TOPIC 12 5 marks

1. A continuous curve has equation $y = f(x)$.

A table of values of x and y for $y = f(x)$ is shown below.

x	0.5	1.75	3	4.25	5.5
y	3.479	6.101	7.448	6.823	5.182

Using the trapezium rule with all the values of y in the given table,

- (a) find an estimate for

$$\int_{0.5}^{5.5} f(x) \, dx$$

giving your answer to one decimal place.

(3)

- (b) Using your answer to part (a) and making your method clear, estimate

$$\int_{0.5}^{5.5} (f(x) + 4x) \, dx$$

(2)

(Total for Question 1 is 5 marks)

Q43**PAST PAPER**

Pearson IAL WMA12 · June 2025 · Q5

P2 · TOPIC 12 6 marks**5:**

x	-2	-0.5	1	2.5	4	5.5	7
y	12	4.243	1.5	0.530	0.188	0.066	0.023

The table above shows corresponding values of x and y for

$$y = 3\left(\frac{1}{2}\right)^x$$

The values of y are given to 3 decimal places as appropriate.

- (a) Using the trapezium rule with all the values of y in the given table, obtain an estimate for

$$\int_{-2}^7 3\left(\frac{1}{2}\right)^x dx$$

giving the answer to one decimal place.

(3)

Using the answer to part (a) and making your method clear, estimate

(b) (i) $\int_{-2}^7 3\left(\frac{1}{2}\right)^{x+2} dx$

(ii) $\int_{-2}^7 (2^{-x} + 2x) dx$

(3)

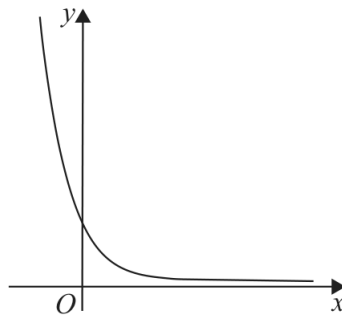
(Total for Question 5 is 6 marks)

Q44**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q3

P2 · TOPIC 12 7 marks

3.

**Figure 1**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of the curve with equation $y = 3 \times 2^{-x}$

The point $P(k, 300\,000)$ lies on the curve.

(a) Use logarithms to find the value of k to 2 decimal places.

(2)

x	-0.5	1	2.5	4.0	5.5	7
y	4.243	1.5	0.530	0.188	0.066	0.023

The table shows corresponding values of x and y for $y = 3 \times 2^{-x}$

The values of y are given to 3 decimal places where appropriate.

(b) (i) Use the trapezium rule, with all the values of y from the table, to find an approximate value, to 2 decimal places, for

$$\int_{-0.5}^7 3 \times 2^{-x} \, dx$$

(3)

(ii) Use your answer to part (b)(i) to estimate

$$\int_{-0.5}^7 2^{-x} \, dx + \int_{-7}^{0.5} 2^x \, dx$$

(2)

(Total for Question 3 is 7 marks)

Q45**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q2

P2 · TOPIC 12 6 marks

2.

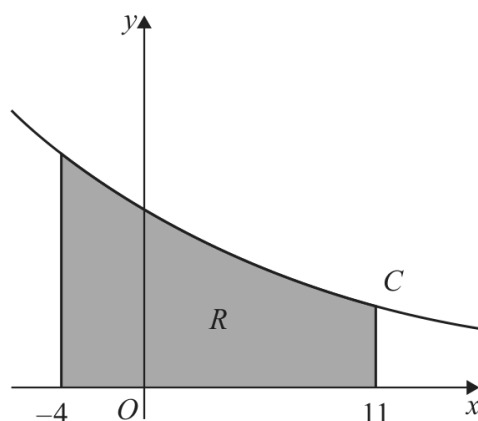
**Figure 1**

Figure 1 shows a sketch of part of the curve C with equation $y = 2^{-0.1x}$

The table below shows corresponding values of x and y for $y = 2^{-0.1x}$

The y values are given to 4 decimal places.

x	-4	-1	2	5	8	11
y	1.3195	1.0718	0.8706	0.7071	0.5743	0.4665

The region R , shown shaded in Figure 1, is bounded by C , the x -axis and the lines with equations $x = -4$ and $x = 11$

- (a) Use the trapezium rule with all the values of y in the table to find an estimate for the area of R . Give the answer to 2 decimal places.

(3)

- (b) State how you would use the trapezium rule to get a more accurate estimate for the true area of R .

(1)

Using the answer to part (a) and showing your working,

- (c) estimate the value of

$$\int_{-4}^{11} 2^{3-0.1x} dx$$

(2)

(Total for Question 2 is 6 marks)

Q46**PAST PAPER**

Pearson IAL WMA12 · June 2026 · Q5

P2 · TOPIC 12 7 marks

5. The table below shows values of x and y for the curve C with equation $y = f(x)$.

x	0.9	1.5	2.1	2.7	3.3	3.9	4.5	5.1
y	1.5	0.48	0.30	0.26	0.26	0.30	0.48	1.5

- (a) Using the trapezium rule with all the y values in the table, find an estimate for the area of the region bounded by C and the x -axis between the lines with equations $x = 0.9$ and $x = 5.1$

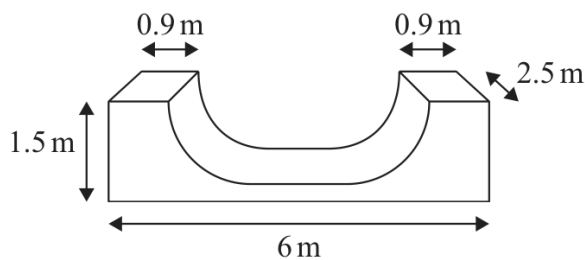
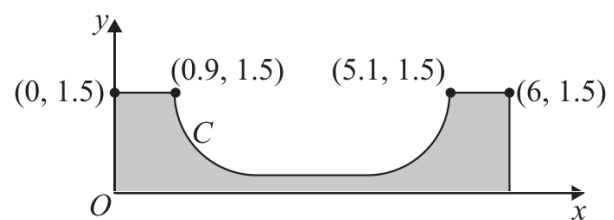
(3)**Figure 1****Figure 2**

Figure 1 shows a sketch of a solid concrete skateboard ramp.

Figure 2 shows a graph of the cross-section of the same ramp.

The ramp

- has a total length of 6 m
- has a platform at each end at a height of 1.5 m and with a length of 0.9 m
- has a curved section C with equation $y = f(x)$, where x and y are in metres
- has a uniform cross-section and is 2.5 m wide

as shown in Figure 1 and/or Figure 2.

- (b) Using the answer to part (a), find an estimate for the volume of concrete in the **whole** of the ramp.

(3)

- (c) State, giving a reason for your answer, whether the answer for part (b) is an overestimate or an underestimate of the true volume of concrete in the whole of the skateboard ramp.

(1)

(Total for Question 5 is 7 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------------|--|
| 01 | Definite integrals | Rewrite, integrate, top – bottom; unknown → equation. |
| 02 | Areas | Meeting points; top – bottom; below the axis is negative. |
| 03 | Trapezium rule | $\frac{h}{2}\{\text{ends} + 2(\text{middle})\}$; convex → over. |

BOTTOM LINE Sketch first • limits from the meeting points • no formula → trapezium rule.

Answers

Integration: Areas and the Trapezium Rule · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · DEFINITE INTEGRALS: LIMITS AND UNKNOWN CONSTANTS

- Q1** Dr Eslam · Evaluate $\left[2x^{\frac{3}{2}} + \frac{2}{x}\right]_1^4 = (16 + \frac{1}{2}) - (2 + 2) = 12.5$
- Q2** Dr Eslam · Find the limit $k^2 + k - 2 = 18 \Rightarrow (k + 5)(k - 4) = 0: k = 4$
- Q3** Practice · Easy (i) 6 (ii) 99
- Q4** Practice · Hard $\int_4^9 (x^{\frac{3}{2}} + x^{-\frac{1}{2}}) dx = \left[\frac{2}{5}x^{\frac{5}{2}} + 2x^{\frac{1}{2}}\right]_4^9 = \frac{432}{5}$
- Q5** Practice · Very Hard $2x^{\frac{5}{2}}: 2q^{\frac{5}{2}}(32 - 1) = 15066 \Rightarrow q^{\frac{5}{2}} = 243: q = 9$
- Q6** Pearson IAL WMA12 · June 2022 · Q7 $A = \frac{1}{2}, B = -28$
- Q7** Pearson IAL WMA12 · January 2026 · Q4 $k = \frac{1}{2} \text{ or } k = 3$

IDEA 02 · AREAS: CURVES, LINES AND BELOW THE AXIS

- Q8** Dr Eslam · Curve and line meet at $x = 1, 4$: $\int_1^4 \{(x+2) - (x^2 - 4x + 6)\} dx = \frac{9}{2}$
- Q9** Dr Eslam · Below the axis $\int_{-2}^2 (x^2 - 4) dx = -\frac{32}{3}$: area $\frac{32}{3}$
- Q10** Practice · Easy (a) $\int_1^2 (9 - x^2) dx$ (b) $\frac{20}{3}$
- Q11** Practice · Hard $\int_0^2 (2x + 3x^2 - 2x^3) dx = 4$
- Q12** Practice · Very Hard meet at $x = 4$ and 9 : $\int_4^9 (-2x^2 + 26x - 72) dx = \frac{125}{3}$
- Q13** Pearson IAL WMA12 · October 2019 · Q8 (i) $4\sqrt{2} - \frac{37}{8}$. (ii) $k = \frac{29}{18}$
- Q14** Pearson IAL WMA12 · January 2020 · Q4 $\frac{20\sqrt{5}}{3}$
- Q15** Pearson IAL WMA12 · January 2022 · Q9 (a) $P = (1-m, m(1-m))$. (b) Proven: $\frac{(1-m)^3}{6}$.
(c) $m = 1 - \frac{1}{\sqrt[3]{2}}$
- Q16** Pearson IAL WMA12 · January 2023 · Q9 (a) $D = (0, 5)$. (b) $x_E = 1, x_F = 3$. (c) $k = \frac{8}{19}$
- Q17** Pearson IAL WMA12 · June 2023 · Q10 (a) $a = 6, b = -84$. (b) $k = 4$. (c) Area $= \frac{1683}{10}$
- Q18** Pearson IAL WMA12 · October 2024 · Q8 (a) $x_P = 1, x_Q = 3$. (b) $\frac{16}{3}$
- Q19** Pearson IAL WMA12 · January 2025 · Q9 (a) $M\left(16, \frac{2304}{5}\right)$. (b) $x_P = 25$. (c) 1312.7
- Q20** Pearson IAL WMA12 · June 2025 · Q3 $\frac{27}{2}$
- Q21** Pearson IAL WMA12 · October 2025 · Q7 (a) $3x^2 - 10x^{3/2} - \frac{k}{2\sqrt{x}} + 28$. (b) Proven. (c) 248
- Q22** Pearson IAL WMA12 · June 2026 · Q8 (a)(i) $(4, -6)$ (ii) 5 (b) $B(8, -9)$ is the minimum:
 $\frac{dy}{dx} = 16a - \frac{3}{2} = 0$ (c) $\frac{25\pi}{2} + 8$

IDEA 03 · THE TRAPEZIUM RULE: ESTIMATE, THEN TRANSFORM

- Q23** Dr Eslam · Trapezium rule
 $h = 0.5$: $y = 1, 1.118, 1.414, 1.803, 2.236$:
 ≈ 2.98
 $3(5.4) + 2(2) = 20.2$
- Q24** Dr Eslam · Use the estimate
 four trapezia of equal width between a and b , tops joining the curve
- Q25** Practice · Easy
 $(a) y = 0, 4.625, 7, 7.875, 8$: 11.75 (b)(i) exact 12: error 2.08% (ii) more (narrower) strips
- Q26** Practice · Medium
 $(a) 5.3364$ (b) exact 5.3544: error $\approx 0.336\%$
- Q27** Practice · Very Hard
 (a) A decreasing exponential curve crossing the y -axis at $(0, 1)$. (b) 19.0. (c) 1.0
- Q28** Pearson IAL WMA12 · October 2019 · Q5
 $(a) 45.6$. (b)(i) 18.2. (b)(ii) -21.6
- Q29** Pearson IAL WMA12 · January 2020 · Q1
 $(a) P(10) = 1.20, P(11.5) = 2.821$. (b) 4.61 kWh
- Q30** Pearson IAL WMA12 · January 2021 · Q7
 $(a) 12.33$. (b) $\frac{2}{5}x^{5/2} - \frac{3}{2}x^2 + 9x + c$. (c) 4.6
- Q31** Pearson IAL WMA12 · June 2021 · Q5
 $(a) 10.10$. (b) Use more strips. (c)(i) 5.05. (c)(ii) 54.30
- Q32** Pearson IAL WMA12 · October 2021 · Q3
 $(a) 11.79$. (b)(i) 23.57. (b)(ii) 17.79
- Q33** Pearson IAL WMA12 · January 2022 · Q1
 $(a) y(1) = 1.075, y(2) = 1.041$. (b) 2.96. (c) 7.04
- Q34** Pearson IAL WMA12 · June 2022 · Q2
 $(a) 4.845$. (b) 25.2%
- Q35** Pearson IAL WMA12 · October 2022 · Q6
 $(a) 10.52$. (b)(i) 6.52. (b)(ii) 10.52
- Q36** Pearson IAL WMA12 · January 2023 · Q1
 6.814
- Q37** Pearson IAL WMA12 · June 2023 · Q1
 $(a) 31.38 \text{ m}^2$. (b) $2.8 \times 10^3 \text{ m}^3$. (c) Under-estimate
- Q38** Pearson IAL WMA12 · October 2023 · Q6
 $(a) y$ -intercept $(0, 5)$ and asymptote $y = 4$.